

Berlin Rent Price Prediction – Final Report

Project: Predicting Cold Rent (Kaltmiete) in Berlin using Machine Learning

Author: Edvin Perfundi

Date: 16.09.2026

Repository: https://github.com/viiniiii/rent_prices_prediction

Website: <https://berlin-rent-price-evaluation.streamlit.app/>

Executive Summary

This report presents the development lifecycle of a machine learning project for predicting apartment cold rent (Kaltmiete) in Berlin, covering data collection, preprocessing, exploratory analysis, feature engineering, model development, evaluation, explainability, ethics and deployment.

Data was collected from Immowelt between 28 December 2025 and 31 January 2026. Of 22,687 raw listings, 15,588 remained after removing invalid, incomplete, and duplicate observations. From 29 collected features, relevant variables were selected and processed using categorical encoding and numerical preprocessing.

Linear Regression, Support Vector Regression, K-Nearest Neighbors, Random Forest, XGBoost, and a Neural Network were evaluated. Although XGBoost achieved the lowest error, Random Forest was selected for its interpretability and tree-disagreement uncertainty proxy. The final model achieved an **R² of 0.9198 and an RMSE of €204.59 on the test set.**

Area and number of rooms were among the strongest predictors, while location, availability, building characteristics, and amenities also contributed. Model explainability was examined using feature importance, SHAP values, and partial dependence analysis. Tree disagreement correlated **0.536 with absolute prediction error**, indicating that greater disagreement tends to be associated with larger errors.

Key limitations include the single scraping period, missing property information, potential representation and measurement bias and weaker performance for extreme rents. The deployed web application provides rent estimates from user-supplied apartment characteristics. Future work will focus on longitudinal data, improved geographical features, and more rigorous uncertainty calibration.

Table of Contents

1	Project Overview	5
1.1	Goal.....	5
1.2	Motivation	5
1.3	Data Source.....	5
2	Data Collection	5
2.1	Legal & Ethical Compliance	5
2.2	Features Collected	5
3	Data Cleaning & Preprocessing.....	7
3.1	Handling Missing Values	7
3.2	Duplicate Removal.....	9
3.3	Outlier Handling	9
3.4	Dataset after cleaning	9
4	Exploratory Data Analysis (EDA)	9
4.1	Univariate Analysis.....	10
4.2	Bivariate Analysis	15
4.3	Correlation Analysis	23
5	Feature Engineering	24
5.1	Encoding	24
5.2	Scaling	24
5.3	Feature Ablation.....	25
6	Modeling.....	25
6.1	Train/Test Split.....	25
6.2	Models Evaluation	25
6.3	Hyperparameter Tuning	26
6.4	Final Model Selection	26
7	Model Evaluation & Uncertainty	27
7.1	Performance	27
7.2	Prediction vs Actual.....	27

7.3	Uncertainty	28
7.4	Error Analysis.....	29
8	Model Explainability	31
8.1	Feature Importance.....	32
8.2	SHAP Values.....	32
8.3	Partial Dependence	33
8.4	Individual Prediction Breakdown	33
9	Ethical Considerations.....	33
9.1	GDPR & Privacy.....	33
9.2	K-Anonymity Check	33
9.3	Bias Discussion	34
9.4	Fairness	34
10	Limitations & Future Work.....	34
11	Conclusion	35
12	Appendices	35

Figure 1:	Histogram of rent prices.	10
Figure 2:	Histogram of apartment's area in square meter.....	10
Figure 3:	Treemap of distribution of number of floors	11
Figure 4:	Distribution of apartments by year of construction	11
Figure 5:	Line chart of number of apartments by year built	11
Figure 6:	Bar chart of condition of the apartment feature.....	12
Figure 7:	Bar chart for each binary categorical feature	12
Figure 8:	Pie chart of heating type.....	13
Figure 9:	Pie chart of electricity source	13
Figure 10:	Pie chart of flooring type	14
Figure 11:	Distribution of apartments based on location	14
Figure 12:	Area vs rent price	15
Figure 13:	Area (up to 220 square meters).....	15
Figure 14:	Box plots of rent prices for each floor number.....	16
Figure 15:	Box plot of rent prices for each availability category.....	16
Figure 16:	Box plots breakdown for each binary feature.....	17
Figure 17:	Interactive scatterplot of property condition (available for all categorical features)..	18

Figure 18: Interactive box plots for different categorizations of apartment's location	19
Figure 19: Median cold rent by district	20
Figure 20: 15 most expensive and affordable districts based on price per square meter	21
Figure 21: Interactive map of Berlin illustrating price per square meter by ZIP code, alongside the distribution of prices per square meter and the relationship between apartment area and rent.....	21
Figure 22: Heat map of number of apartments grouped by construction year and rent prices ..	22
Figure 23: Scatterplot of apartments area based on year of construction (number of rooms encoded using color)	22
Figure 24: Apartments by number of rooms across the decades.....	23
Figure 25: Scatterplot matrix between rent price, area, number of rooms and number of floor	23
Figure 26: Matrix of correlation between numerical features	24
Figure 27: Matrix of correlations stronger than 0.30.....	24
Figure 28: Radar charts for the main models' performance.....	26
Figure 29: Elbow method chart for number of trees	26
Figure 30: Elbow method chart for tree's depth	26
Figure 31: Scatterplot of relation between predicted and actual rent prices	27
Figure 32: Scatterplot with confidence intervals for predicted vs actual values	28
Figure 33: A breakdown of a single Random Forest prediction.....	28
Figure 34: Error distribution.....	29
Figure 35: Predicted vs residuals graph	29
Figure 36: Mean Absolute Error by district	29
Figure 37: Error distribution by district.....	29
Figure 38: Feature importance ranking.....	32
Figure 39: SHAP values for overall prediction	32
Figure 40: Partial dependence of predicted rent and area.....	33
 Table 1: Metrics for each trained model	 25
Table 2: Metrics for each value of 'property_condition' feature	30
Table 3: Metrics for each value of 'heating_type' feature	31
Table 4: Metrics for each value of 'rooms' feature	31

1 Project Overview

1.1 Goal

To develop a machine learning model that accurately predicts the cold rent (Kaltmiete) of apartments in Berlin based on property features such as size, number of rooms, location, amenities and building characteristics.

1.2 Motivation

Warmmiete (rent including utilities) is not consistently available across listings, making Kaltmiete the more reliable target variable. Accurate rent prediction can assist renters, landlords and analysts in understanding market dynamics.

1.3 Data Source

- **Source:** Immowelt (German real estate marketplace)
- **Reason:** Rich feature set per listing, high volume of new listings daily.
- **Scraping Period:** 28/12/2025 – 31/01/2026
- **Raw Listings Scraped:** 22,687
- **Final Listings After Cleaning:** 15,588 (modeling dataset)
- **Raw Features Collected:** 29
- **Final Model Features:** 15 (before one-hot encoding)

2 Data Collection

A custom scraping script was written to collect data from the website.

2.1 Legal & Ethical Compliance

- No APIs were available and scraping was performed respecting robots.txt.
- GDPR compliance: No personal data (names, phone numbers) was scraped.
- Only publicly available listing data was collected.
- `time.sleep()` was used to avoid overloading the server.
- No copyright-protected content (articles, images) was scraped.

2.2 Features Collected

1. Listing URL – unique identifier
2. Cold rent (Kaltmiete) – target variable

3. Warm rent (Warmmiete) – dropped due to data leakage
4. Deposit (Kaution) – dropped due to data leakage
5. Area (m²)
6. Number of rooms
7. Floor
8. Free from (availability)
9. Location (address, district, ZIP)
10. Has balcony
11. Has terrace
12. Has garden
13. Has elevator
14. Has parking
15. Has basement (Keller)
16. Barrier-free
17. Has built-in kitchen (Einbauküche)
18. Has bathtub
19. Has shower
20. Flooring type
21. Energy source
22. Heating type
23. Property condition
24. Year built
25. Energy certificate
26. Energy demand
27. Schufa check required
28. Number of images posted
29. Scraped at (timestamp)

3 Data Cleaning & Preprocessing

This section contains the most labor-heavy work that requires detailed inspection, domain knowledge and application of different cleaning and preprocessing methods.

3.1 Handling Missing Values

Almost all collected features had missing data, hence the reason why cleaning the data needs a lot of time.

Boolean Features

For all Boolean features, it was assumed that the feature was not present in case of absence of values, since the scraper works that way that sets the values to 'True' when the feature is mentioned in the apartment listing. For example, the word 'balkon' (in German) not being mentioned anywhere for a listing would cause the scraper to put the value to 'False' for the feature named 'has_balcony'. The Boolean features are the following: 'has_terrace', 'has_garten', 'elevator', 'parking', 'has_basement', 'is_barrier-free', 'has_built-in_kitchen', 'has_bathtub', 'has_shower', 'has_balkon'.

Cold Rent (Target feature)

This is our main feature and for that reason it shouldn't have any missing values. After removing all invalid listing, there were only 398 missing values (1.75%), which made the cleaning much relaxed. Having a large dataset dropping this amount of values won't damage it and it is much more recommendable than inputting the values and possibly creating additional noise.

Area and Rooms

These 2 features are the features with the most impact on price. After dropping cold_rent nulls very few samples remained with missing area and those did happen to not have most of the other features making them invalidate. Initially with 1031 samples with missing area and 1008 with missing rooms feature. We dropped the rows with missing area since it's the main predictor. Missing rooms dropped to 161 samples and after inspection, they were valid, but it was decided to still drop them (0.75% of data) to avoid noise.

Location

This is also a very important predictor with 326 missing (1.44%). During inspection it was revealed that all the listings were in Berlin, but the values were listing either due to scraper failure or listing formatting. The decision was initially postponed and later we decided to drop 13 rows with missing ZIP and map the remaining manually. Final missing location handled by dropping rows (lost ~2% of data).

Floor

Number of apartment floor was one of the most complex features to deal with due to multiple formats, for instance: 5. Geschoss, EG, 9 Geschosse, fractions like Geschoss 3/5, negative values. For this reason, a 'format_floor' function was created to do the following:

- Extracts floor number from X. Geschoss and Geschoss X/Y.
- EG → 0.
- 9 Geschosse (total floors only) → NaN.
- Negative values: kept -1 as basement, converted others to positive.

The continuous floor feature with NaNs was preserved.

Warm Rent and Deposit

It was a good idea to drop both features because they are directly linked to cold rent and that would cause data leakage.

Free From (Availability)

Like number of apartment floor, the availability of apartments had many formats such as dates, sofort, nach Absprache, NaN. The feature was transformed into 3 categories: 'immediately' that would imply that the tenant can immediately rent and move-in into the apartment, 'not_immediately' and 'unknown' which implies that the information regarding the move-in date is unclear or simply missing.

This kind of approach instead of using more information from exact dates due to the missing information about the date posted of listings and therefore a move-in availability in number of days couldn't be derived.

Categorical Features

These features include several values and usually seem to carry a lot of information. The features include the following: 'flooring_type', 'energy_source', 'heating_type' and 'property_condition'. Missing values replaced with "Unknown", which also was proved to carry a lot of information, such as negative association with the predicted rent price.

Year Built

The 4-digit year was extracted and values less than 1700 were set to 'NaN', leaving it with values ranged from 1765 to 2026.

Number of Images Posted

20% of its values were missing and no reliable imputation or information was extracted out of this feature hence the reason why it was dropped entirely.

3.2 Duplicate Removal

There were many duplicate samples in the dataset, but initially no exact duplicates were found due to the 'scraped_at' value causing every sample to be unique. Being unable to derive any information out of this feature, it was decided to drop it later. We checked duplicates excluding URL and scraped_at and found 20% duplicates (similar listings reposted). Duplicates were removed and the most recent listings were kept referring to the 'scraped_at' feature. Additional checks for near duplicates (excluding one feature) were made and other relatively identical samples were further removed. The final duplicated dataset contained 15,803 samples out of 20,932 before deduplication.

3.3 Outlier Handling

The dataset contains many continuous features which also tend to have many outliers. A lot of inspecting was needed to identify these outliers and validate if those values are valid.

Cold Rent:

Values under 300 € were inspected and some of them were caused due to formatting issues (e.g., 1.349 → 1349). Some ambiguous listings were dropped, and extreme values were dropped which included top 1% (≈ 160 samples).

Area:

Values under 10 m² were inspected and concluded as invalid and therefore dropped. Extremely large values included luxury listings with very high area were kept.

Rooms, Floor, Property Condition and Year Built:

All these continuous features were inspected and they were all considered to be valid.

3.4 Dataset after cleaning

The final dataset has:

- **Samples:** 15,621
- **Features:** 27
- **Samples dropped:** 7066

Number of features didn't initially change because of the derived features (from breakdown of location feature). Further changes will be made during the later stages of analysis.

4 Exploratory Data Analysis (EDA)

This section includes a lot of visualizations and produces the final dataset after deeper feature analysis.

4.1 Univariate Analysis

This section includes inspecting and visualizing each feature individually to break it down and gain detailed information out of it, potentially identifying more invalid features and performing further formatting and cleaning on the features.

Rent Distribution

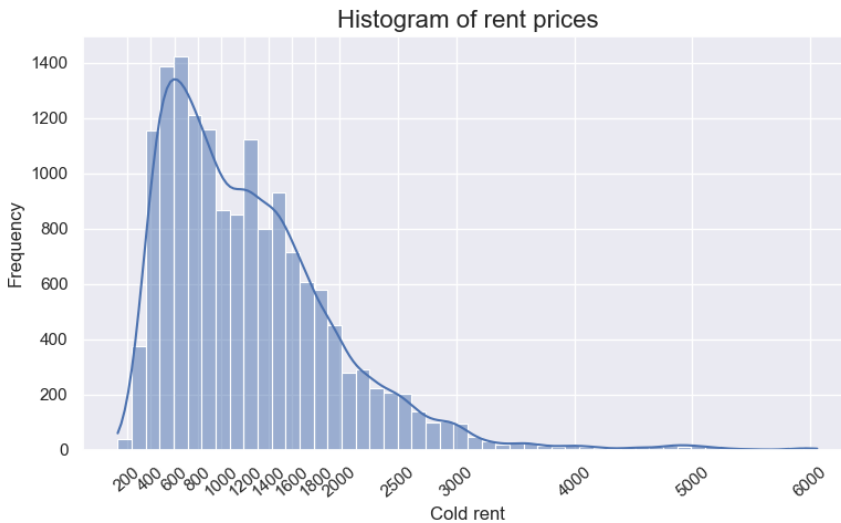


Figure 1: Histogram of rent prices.

Referring to figure 1, we can clearly notice that the distribution of rent prices is strongly right-skewed, with most rents being between 500 € and 2,000 €. The median is 1,100 € and there are common spikes at 750 €, 1,200 €, 1,500 €.

Area

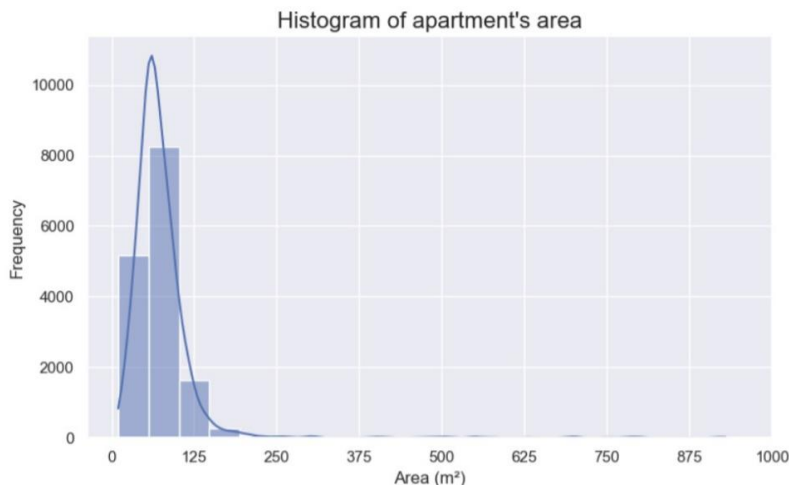


Figure 2: Histogram of apartment's area in square meter.

The apartments seem to have a size distribution centered around 50–70 m². Mean value is 78.65 m² and the median: 65.8 m². The highest value is 5,251 m² (luxury/commercial).

Rooms

Most common: 2 rooms, followed by 3, 1, 4. Distributed from 1 to 35 rooms, but values above 7 are almost inexistent and removed.

Floor

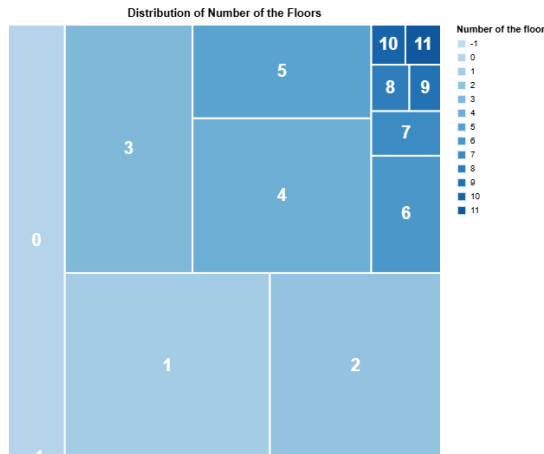


Figure 3: Treemap of distribution of number of floors

Most apartments are on 1st floor, with 4th, 2nd, 3rd being also popular.

Year of construction

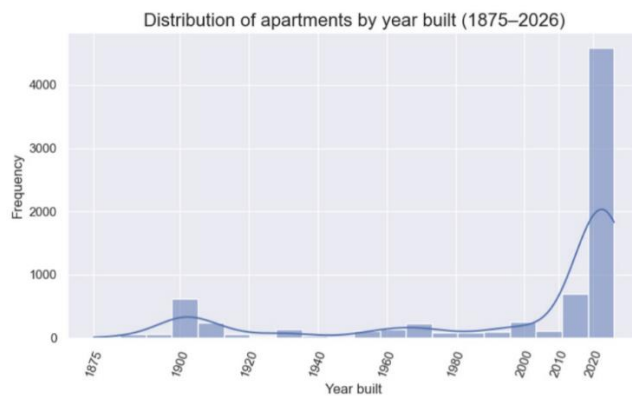


Figure 4: Distribution of apartments by year of construction

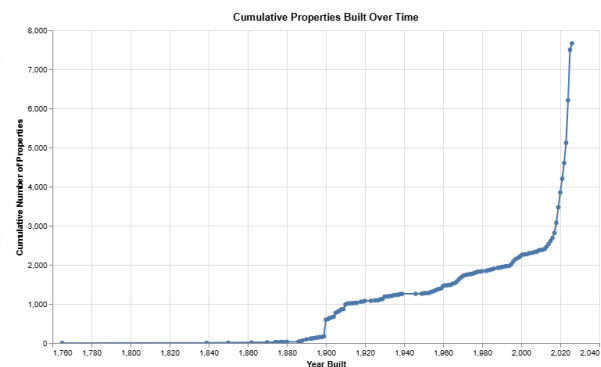


Figure 5: Line chart of number of apartments by year built

There are some spikes in 1900, 1910, 1930 (rounded years) and the periods corresponding to the two World Wars appear with visible dips in the number of observed apartments by construction year. The number of apartments start to grow steadily and after 2010 the number of apartments explodes.

Property Condition

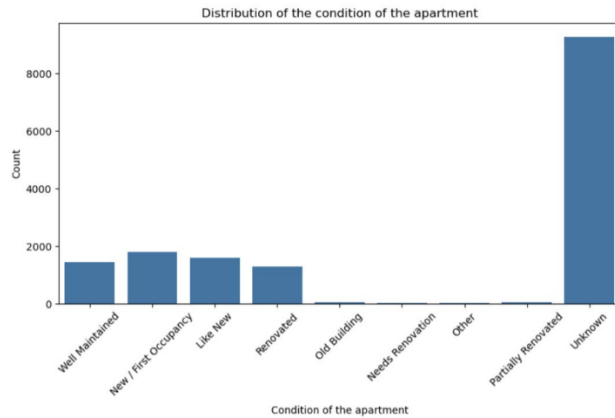


Figure 6: Bar chart of condition of the apartment feature.

This is the categorical feature with the most values, precisely 9, with 'Unknown' being majority (9,284 values). The other values include 'New/First Occupancy' (1,796 values), 'Like New' (1,609 values), 'Well Maintained' (1,453 values), 'Renovated' (1,306 values), 'Old Building' (56 values), 'Partially Renovated' (44 values), 'Needs Renovation' (38 values), 'Other' (35 values).

For this feature we notice that the 4 values with positive association tend to be a lot larger in numbers than the 4 values with negative association. The reason for this could be that the renters avoid putting negative descriptions and instead they leave it empty, which is the reason for the large number of 'Unknown' values. This is not an assumption, but rather our models were able to capture this relation. This relation was discovered even before training the ML models during EDA visualizations, which clearly showed each category in clusters with the good categories being associated with higher prices for the same apartment sizes.

Amenities

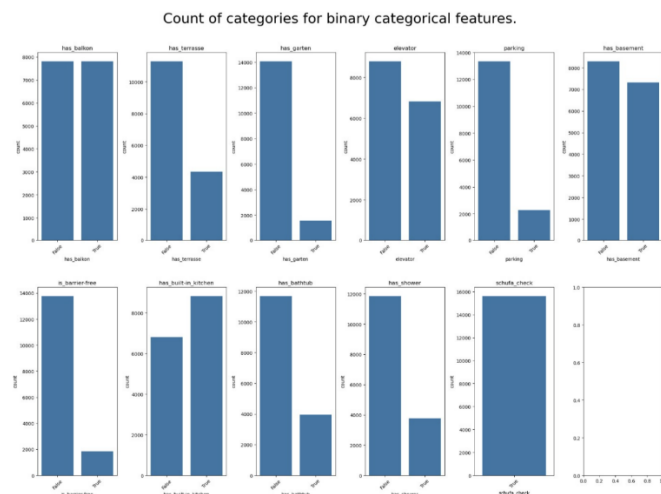


Figure 7: Bar chart for each binary categorical feature

There are 11 binary categorical features. Schufa check feature is invalid since it has only one value, for that reason it was dropped. Some of the features seem very balanced like ‘elevator’, ‘has_basement’ and especially ‘has_balkon’ being equally distributed. ‘has_built-in_kitchen’ is the only feature with ‘True’ as most values. Many of the features have relatively very little occurrences and that is most likely an underreported amount which leads to noise. That causes most of the features to have unexpected or unclear effects. After many analyses only 3 out of 11 binary categorical features made it to the final model.

Heating type

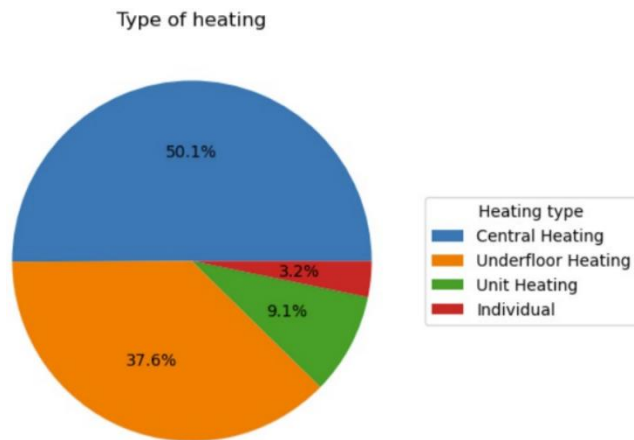


Figure 8: Pie chart of heating type

Unknown makes 42.8% of the dataset when it comes to Heating type values. If we look at the distribution of values without the Unknown category, by also referring to Figure 8 we notice that ‘Central Heating’ is very popular, making up half of the remaining values, followed by ‘Underfloor Heating’, which seems to be also in a considerable amount. ‘Unit Heating’ and ‘Individual’ are less popular but still present in a sufficient number of listings to capture their relation to rent price.

Electricity source

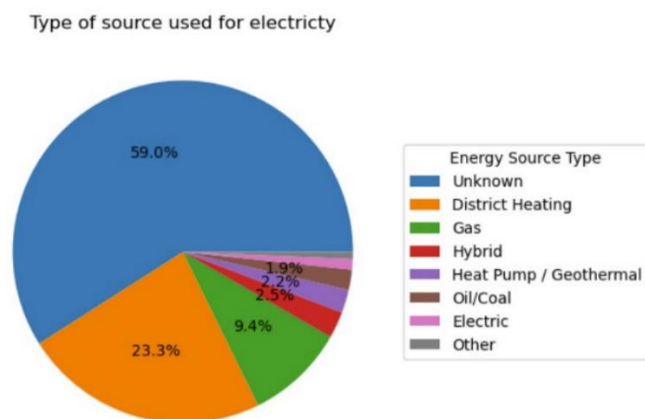


Figure 9: Pie chart of electricity source

Figure 11: Distribution of apartments based on location

District is a feature derived from the full location using the district part of the full value or when missing zip codes were used to derive the values. Sometimes arbitrary positions were needed to decide in which district the apartments fitted the best. Despite the very large number of districts in Berlin, there seems to be a good distribution with even unpopular districts having a sufficient number of samples with even the least popular districts being present in more than 30 samples.

4.2 Bivariate Analysis

Bivariate analysis includes looking at more than 1 feature simultaneously, usually to observe the relationship between them. It is very important to include this analysis for our target variable to see how each feature impacts it. We also get to look at how other features impact each other.

Rent vs Area

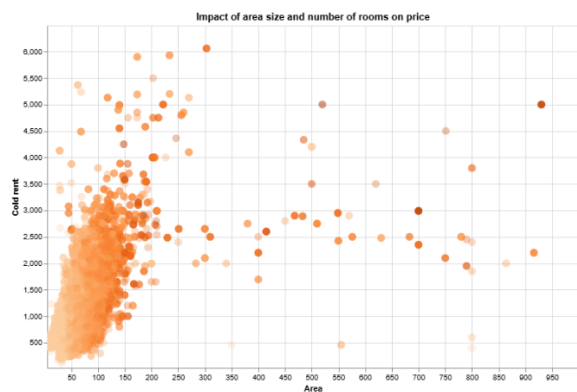


Figure 12: Area vs rent price



Figure 13: Area (up to 220 square meters)

The two scatterplots' graphs above show the relationship between area and price, which is concluded to be the strongest correlation. A strong positive correlation seems to be present, for apartments up to 250 square meters. For larger apartments area has relatively a weaker effect with other features impacting the prices. If we model a regression with only the area, we will have the following:

$$\text{cold_rent} = 146.79 + 15.23 * \text{area}$$

That implies that each additional m² increases rent by 15.23 € on average.

Rent vs Rooms

There is a positive correlation, but slightly weaker than area.

Rent vs Floor



Figure 14: Box plots of rent prices for each floor number

Number of floors seems to have an unclear effect on the rent price. By referring to picture number 14, it's hard to tell what the exact effect of floors on the rent prices is or if there is a difference between floors when it comes to the prices. By performing a statistical test or more precisely a Kruskal-Wallis test there was proved to be a significant difference between floors ($p < 0.05$). Ground floor appears to be visually significantly different, but even after removing it, the difference between floors is still significant.

Rent vs Availability

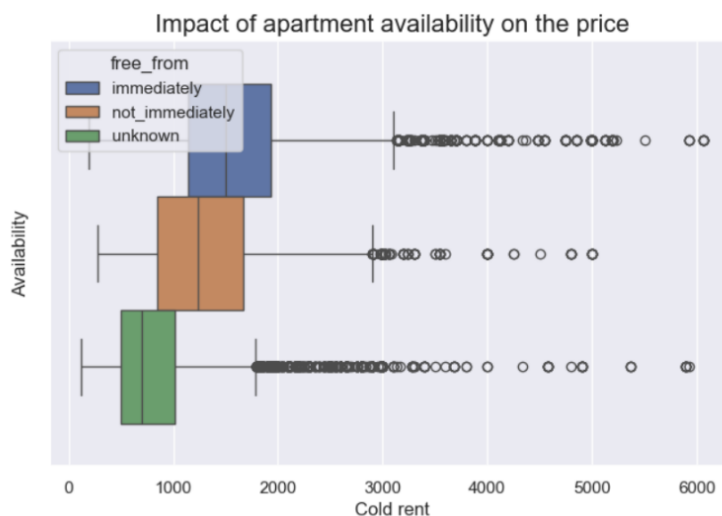


Figure 15: Box plot of rent prices for each availability category

By referring to figure 15, we can clearly notice that apartments with an immediate move-in date are associated with higher rent, with a defined not immediate date lower rent and when there is no information (unknown category), then we have relatively the lowest prices.

Rent vs Amenities

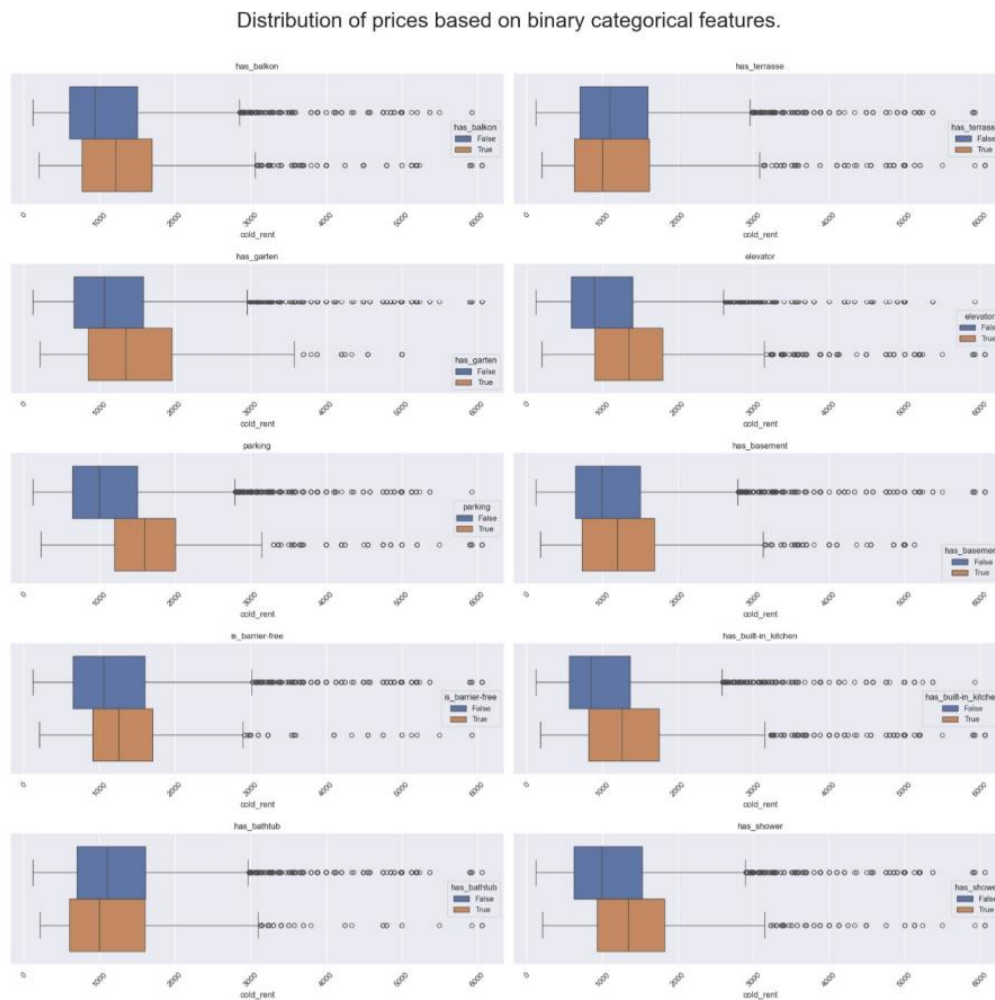


Figure 16: Box plots breakdown for each binary feature

From the boxplots we notice that features like shower, park, elevator, built-in kitchen are clearly associated with higher rent. Terrace and bathtub showed unexpected negative association. The rest had an unclear effect and were later dropped for better performance and model useability.

Rent vs Property Condition

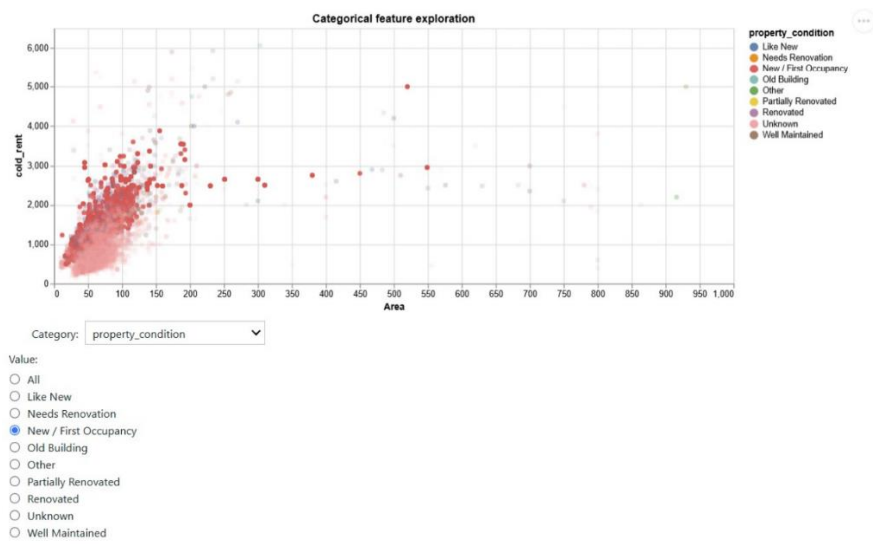


Figure 17: Interactive scatterplot of property condition (available for all categorical features)

The interactive graph in figure 17 allows us to interact with every categorical feature by selecting certain values to inspect them in detail. For property conditions ‘New/First Occupancy’, ‘Renovated’ and other positive values were clustered on the upper left part of the graph meaning that they have higher prices for the same apartment sizes. Needs Renovation had unexpectedly higher than average and the value with the lowest prices was ‘Unknown’.

Rent vs Flooring

By using the graph in picture 17 for the flooring feature, we notice that features like wood, stone, tiles are associated with higher rent and carpet, vinyl, laminate with lower rent.

Rent vs Energy Source

When it comes to energy source the effect among features is hard to detect, but hybrid, electric are clearly associated with higher prices, while unknown values are associated with much lower prices.

Rent vs Heating Type

For heating type underfloor heating is associated with the highest prices, it clearly makes a difference by boosting the apartment prices. Individual heating seems to also be correlated with high price and unit heating seems to have the lowest prices among these 3.

Rent vs District

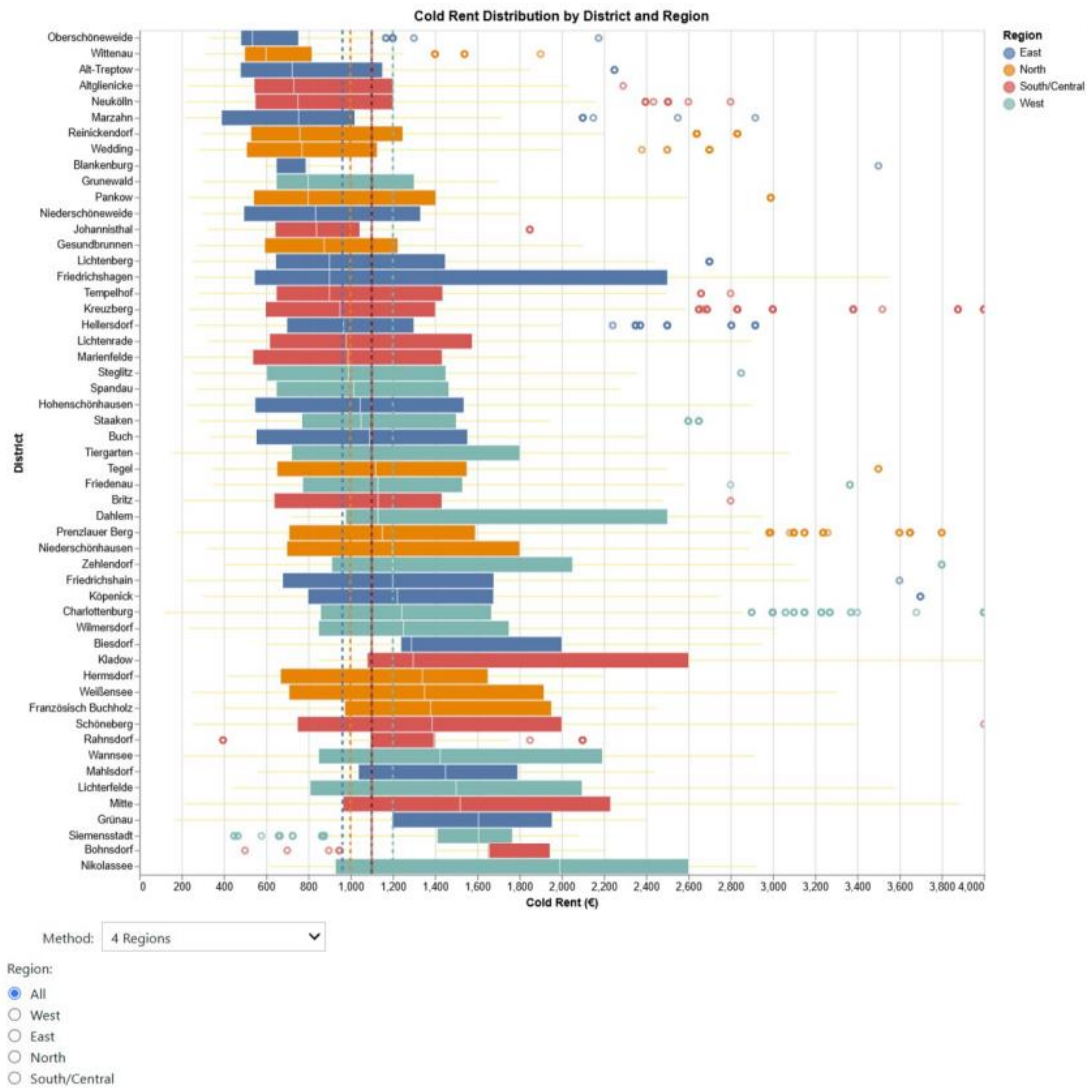


Figure 18: Interactive box plots for different categorizations of apartment's location

After splitting Berlin into 4 regions, west and south of Berlin were associated with higher prices than east and north. If we split into east and west, there is no clear difference and if we split based on distance from city center, regions closer to the city center are associated with higher prices.

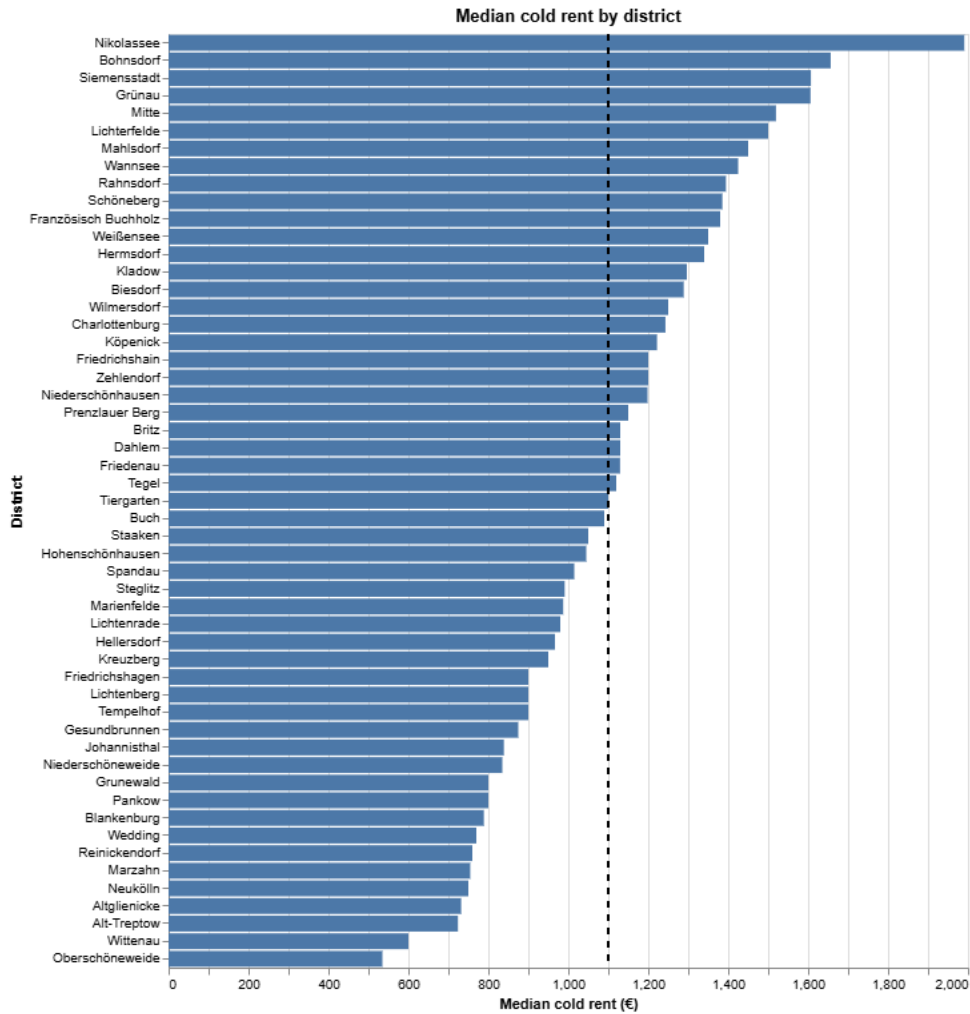


Figure 19: Median cold rent by district

Regions with the most expensive apartments include Nikolassee, Bohnsdorf, Grünau, Mitte, Lichterfelde, etc. Oberschöneweide, Wittenau and Alt-Treptow rank the lowest with median rent prices across 500 to 700 euros, while the overall median rent is around 1100 euros.

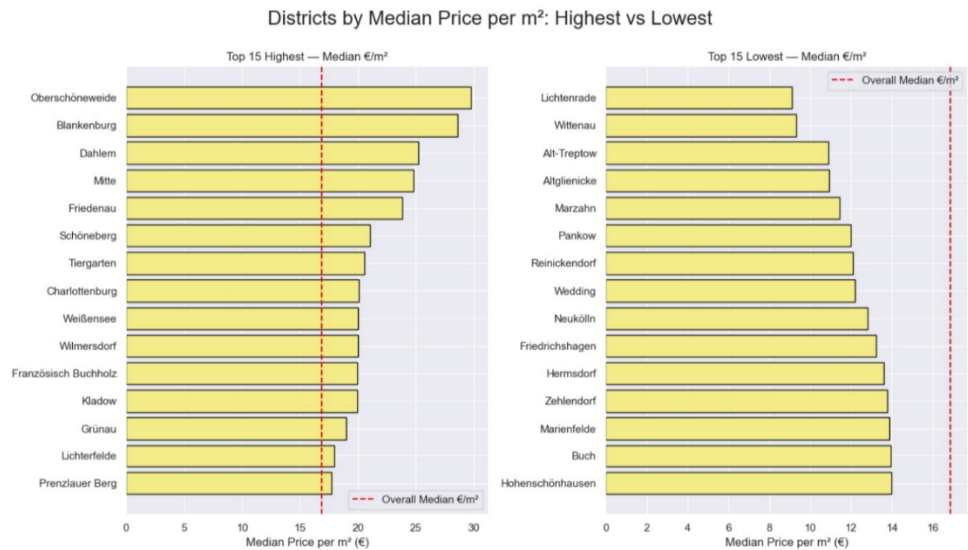


Figure 20: 15 most expensive and affordable districts based on price per square meter

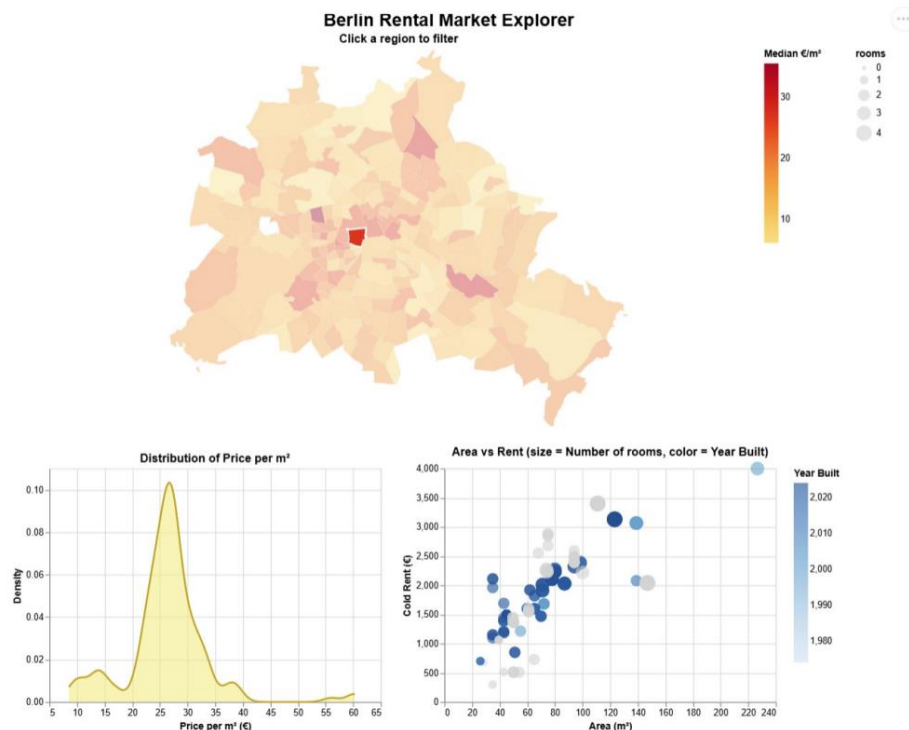


Figure 21: Interactive map of Berlin illustrating price per square meter by ZIP code, alongside the distribution of prices per square meter and the relationship between apartment area and rent.

Median price and price itself are not sufficient to compare different districts due to the possibility of certain districts being associated with larger properties, leading to higher prices because of area rather than the district itself. For this reason, it is also suggested to compare the districts based on price per square meter, which is conducted in figure 20. Oberschöneweide had the lowest median price, but it has the highest median price per square meter. Blankenburg, Dahlem and popular central districts like Mitte, Friedenau, Tiergarten and Charlottenburg lead as the

most expensive districts in terms of price per square meter. Neukölln, Prenzlauer Berg, Lichtenberg, Kreuzberg have most low-price listings. Alt-Treptow remains as a very affordable district with lower overall median and price per square meter median. Overall median is around 16 €/m².

Rent vs Year of construction

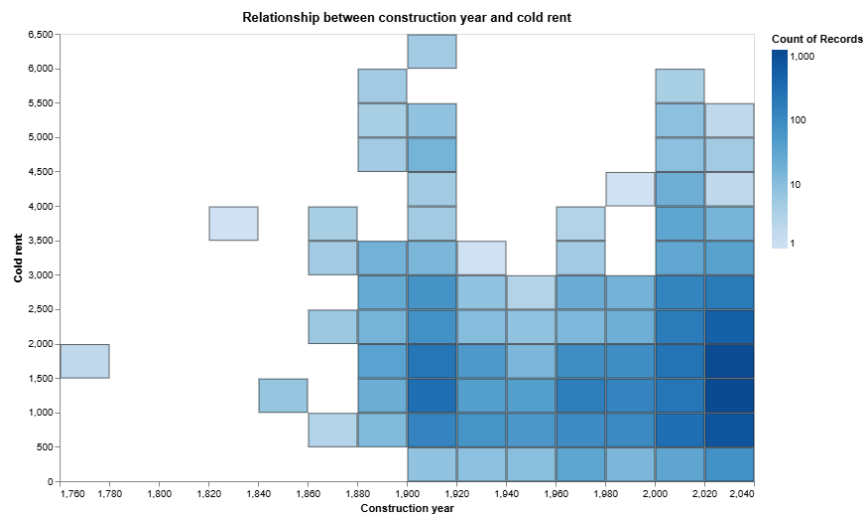


Figure 22: Heat map of number of apartments grouped by construction year and rent prices

There is no clear relationship between price and construction year. The only clear takeaway is that during certain periods such as during 1880-1900 and after 2000 more apartments are constructed and there is a wider range of prices for these periods.



Figure 23: Scatterplot of apartments area based on year of construction (number of rooms encoded using color)

There is also no clear association between year of construction and apartment's size with the same periods mentioned above having a wider variety of values and number of rooms.

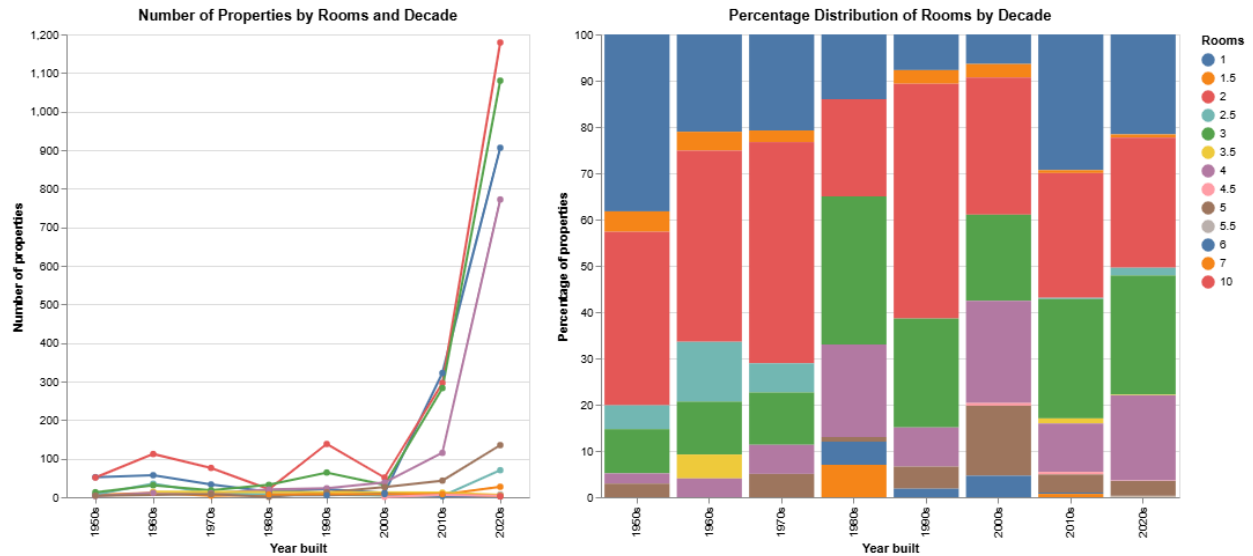


Figure 24: Apartments by number of rooms across the decades

There is a change of trend in number of rooms. Initially 1 and 2 room apartments used to take about $\frac{3}{4}$ of the market but through the years apartments with more rooms like 3, 4 and 5 room apartments were constructed and taking half over half of the market.

4.3 Correlation Analysis

Target variable is not the only feature who interacts with the rest of the features. Other features can also interact with each other, for instance, logically bigger apartments tend to have more rooms so we can expect that area and number of rooms to have a positive correlation. To discover other less visible and logical relationships among variables we perform the correlation analysis.

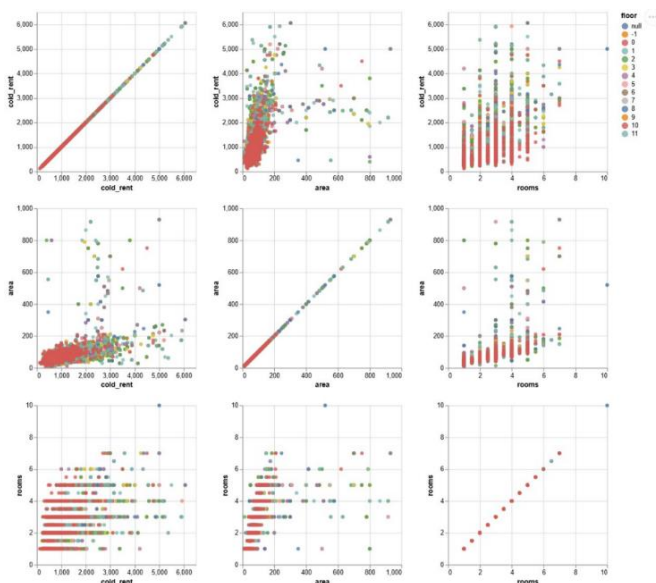


Figure 25: Scatterplot matrix between rent price, area, number of rooms and number of floor

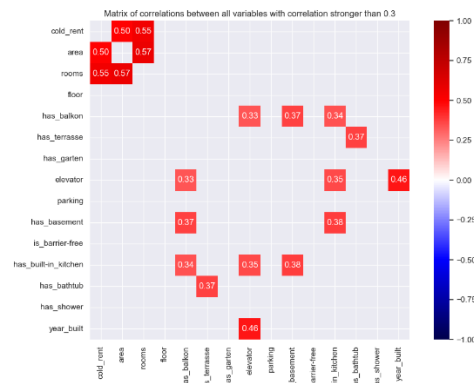
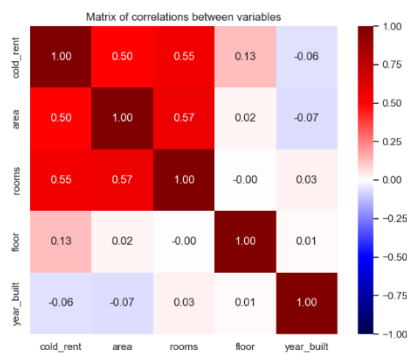


Figure 26: Matrix of correlation between numerical features Figure 27: Matrix of correlations stronger than 0.30

By referring to figure 25, 26 and 27 we notice many correlations between features. The strongest correlations include strong positive between area and rent, rooms and rent, rooms and area. We also notice (figure 25) that ground floor apartments tend to have lower prices so we could also suggest that floor number has an impact on the price, but a very weakly positive one, which is also confirmed by the matrix in figure 26. There is a very weak negative correlation between year-built and rent. One possible explanation is that older apartments in this dataset tend to have larger areas, which is suggested by the negative correlation between year built and area.

Amenity correlations include balcony with elevator, balcony with basement, balcony with built-in kitchen, terrace with bathtub, elevator with built-in kitchen, elevator with year built. Most of these have very strong econometric interpretation, for example, apartments with elevators can have balconies since they are on higher floors, compared to apartments on the first floor who can't have any. There is also an interesting correlation between year-built and elevator having a very strong correlation, which suggests that newer apartments are very likely to have an elevator compared to the apartments constructed earlier.

5 Feature Engineering

This section includes transforming the features for training machine learning models. Feature engineering includes things like encoding, scaling and feature ablation.

5.1 Encoding

- Categorical features: one-hot encoding.
- Final training dataset shape: (15,588, 93).

5.2 Scaling

- Numerical features: Standard Scaling.

5.3 Feature Ablation

- Trained models with all features, -2 features (terrace, bathtub), -7 features (also balcony, garden, parking, basement, barrier-free).
- Random Forest and XGBoost performed better without noisy features.

6 Modeling

Several regression models were trained and evaluated using RMSE and R^2 . XGBoost achieved the strongest predictive performance, followed by Random Forest. Neural Network models also achieved R^2 values above 0.90, while KNN, SVR and Linear Regression performed less well. Feature ablation showed that Random Forest and XGBoost generally benefited from removing features that had been identified as noisy during the exploratory analysis.

6.1 Train/Test Split

Dataset is split 80% training, 20% testing, which would translate to 12470 samples for training and 3118 for testing.

6.2 Models Evaluation

Several machine learning models were trained and evaluated using metrics such as R^2 , RMSE, MAE, etc. The trained models include Linear Regression, SVR, KNN, Random Forest, XGBoost and Neural Network.

Model	RMSE	R^2
XGBoost (2 nd version)	177.29	0.9398
XGBoost	184.13	0.9350
Random Forest (3 rd version)	204.44	0.9199
Random Forest (2 nd version)	206.31	0.9184
Random Forest	209.19	0.9161
Neural Network	222.38	0.9082
Neural Network (2 nd version)	221.42	0.9061
KNN	265.52	0.8649
SVR	408.33	0.6806
Linear Regression	408.98	0.6796

Table 1: Metrics for each trained model (Random Forest 3rd version XGBoost 2nd version were trained without the seven features we decided to remove and Random Forest 2nd version and Neural Network 2nd version were trained without 2 features – has_terrace and has_bathtub).

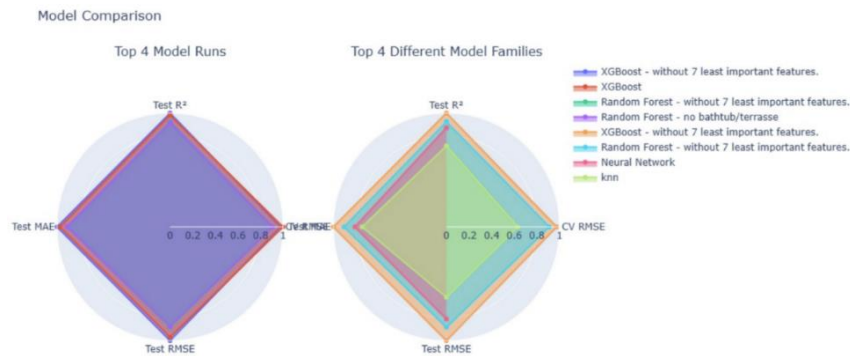


Figure 28: Radar charts for the main models' performance

Referring to table 1 and figure 28, the best performing model is 2nd version of XGBoost, followed by the normal XGBoost. Random Forest seems to be performing almost as good as XGBoost. Neural Network has also a good performance with an R square of around 0.9. SVR and Linear regression perform very poorly in this dataset. What is to be noticed is that the models perform better when trained without the features that were evaluated to be noisy and without a clear effect, for that reason it was decided to favor training them without these features. Only the neural network loses accuracy when trained with less feature, these might imply that can handle noisy features better than other models.

6.3 Hyperparameter Tuning

- Grid Search with 5-fold cross-validation.
- Best Random Forest: 200 estimators, depth 40.
- Best XGBoost: higher values but risk of overfitting.

6.4 Final Model Selection

Random Forest was chosen over XGBoost despite slightly lower R^2 because it natively provides uncertainty estimates via tree disagreement. XGBoost Quantile Regression performed poorly, hence the reason Random Forest was favored. Random Forest offers a good balance of accuracy, interpretability, and uncertainty.

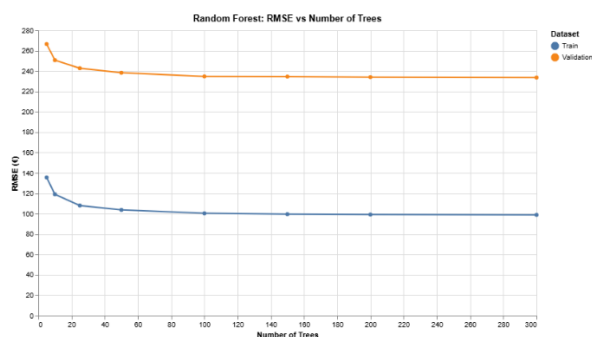


Figure 29: Elbow method chart for number of trees

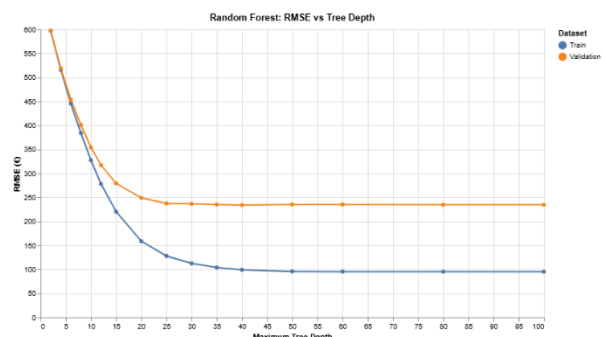


Figure 30: Elbow method chart for tree's depth

To avoid overfitting it is a good idea to see how the model's performance changes for different parameter values. For example, the deeper each individual tree used during training is, the better the model will perform on the training data and potentially also on the test data. However, this does not mean that the depth should be increased indefinitely because this would lead to overfitting the model and reducing its ability to generalize to unseen data. That is why by plotting and examining how performance changes for different parameters we aim to identify parameter values beyond which further increases provide little improvement. This region, from which the parameter values can be selected, can be considered an 'elbow' in the performance curve. By observing the graphs 200 tree estimators and a maximum tree depth of 40 were selected.

7 Model Evaluation & Uncertainty

Further evaluations are needed after selecting the final model for several reasons. For example, it is necessary to see where the model struggles and where it performs very well, rather than just reading some plain general numbers such as testing RMSE or R^2 .

7.1 Performance

- **RMSE:** 204.59 €
- **R^2 :** 0.9198
- **MAE:** 124,74
- **Overfitting gap R^2 :** 0.0635

7.2 Prediction vs Actual

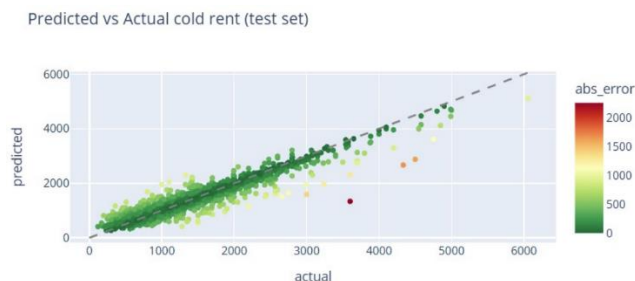


Figure 31: Scatterplot of relation between predicted and actual rent prices

The scatter plot above shows the relation between the actual and predicted values. The model seems to be doing very good, with very few outliers, specifically the one with id 2811 with a huge error of 2261. We can notice that for small values (under 800) the model seems to overpredict and for large values the model seems to underpredict. Especially for very large values, which are all below the line meaning that the model will always underpredict apartments with rent more than 3500 euros.

Predictions with ± 1 std uncertainty band

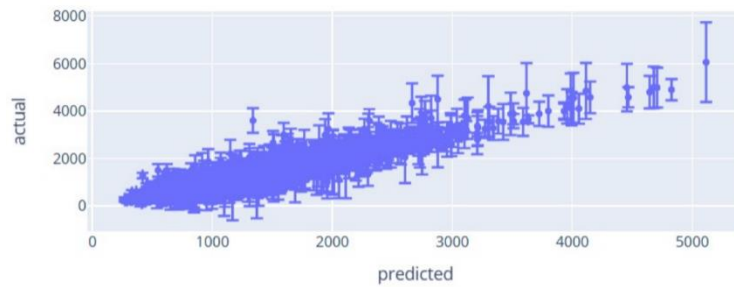


Figure 32: Scatterplot with confidence intervals for predicted vs actual values

The graph above, similar to the previous one, shows the points, using the same variables, but this time using uncertainty to encode the size of segments.

We notice that there is a correlation between the rent price and uncertainty, that could be because we had a right-skewed distribution of apartment prices and outliers tend to have a higher uncertainty, hence the very expensive apartments having the highest level of uncertainty.

A correlation of 0.536 indicates a moderate-to-strong positive relationship between tree disagreement (`pred_std`) and actual prediction error (`abs_error`).

Our Random Forest knows when it doesn't know. When the individual trees in the ensemble disagree strongly on a property's rent, the final averaged prediction is much more likely to be wrong. This is very useful because we don't want to show the user a single value for each prediction but rather use it to communicate whether the user can trust the prediction or not.

7.3 Uncertainty

Listing #0: 200 individual tree predictions

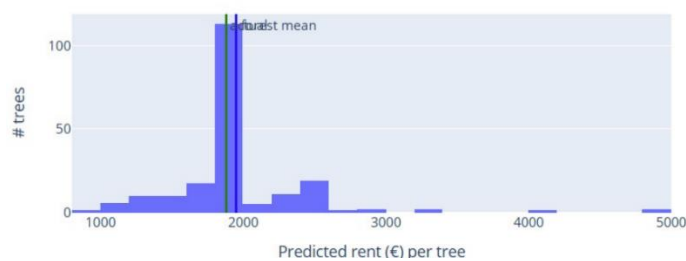


Figure 33: A breakdown of a single Random Forest prediction

- Tree disagreement (`pred_std`) correlates with absolute error: **0.536** (moderate-strong).
- Higher uncertainty for expensive apartments (right-skewed distribution).
- Uncertainty exposed as low/medium/high confidence.

Our Random Forest works in a way that out of n number of trees, each individual tree makes its own prediction and the final prediction is obtained by aggregating the predictions of all trees, hence the reason RandomForest is an ensemble model. We trained our Random Forest with 200 estimators, meaning that each observation receives 200 individual predictions, one from each tree.

The graph above shows us the breakdown of one of the predictions. The bar chart has grouped the predictions into ranges and shows how many trees' predictions are contained in that range. The graph initially suggests a relatively high degree of variation among the individual tree predictions, as the predictions range from 0 to 3200 euros, but most of the trees, precisely 113 trees, are in the correct range. Also, the absolute error is only 69 euros for a price of 1881. That is an error of 3.1% which makes it a very accurate prediction.

7.4 Error Analysis

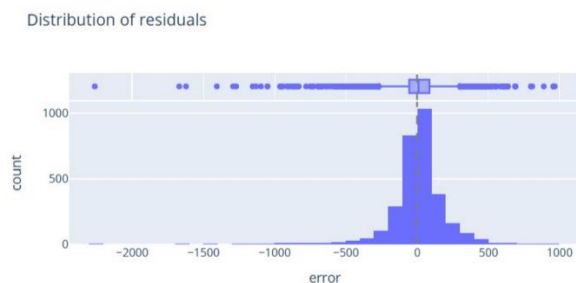


Figure 34: Error distribution

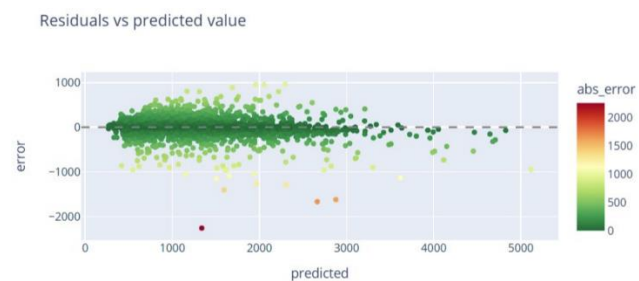


Figure 35: Predicted vs residuals graph

The histogram and box plot show the distribution of errors. A well-performing regression model should have residuals centered around zero and approximately normally distributed. Our model demonstrates these characteristics, suggesting that its predictions are generally unbiased.

The graph above shows the relation between residuals and predicted values. Most of the values are concentrated around 0 and the outliers here are between predicted values of 1000 and 3000. This shows that the models tend to squeeze the very large values into this range, hence being the reason for the very large absolute errors.

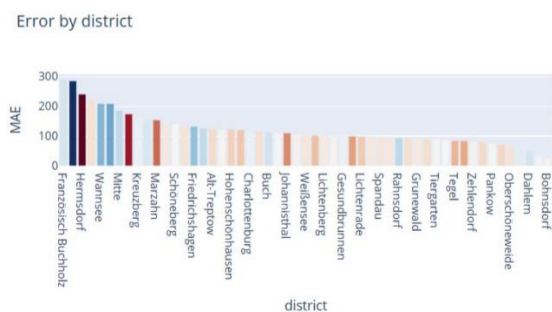


Figure 36: Mean Absolute Error by district

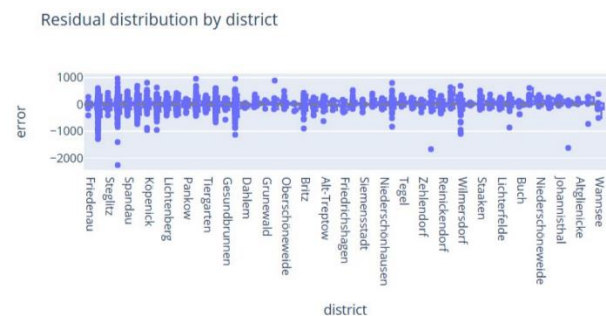


Figure 37: Error distribution by district

The first graph shows districts associated with the highest RMSE errors. Surprisingly, there is not a very strong correlation between sample size and error. The leading districts seem to be both districts with small samples as well as with large samples, with districts with small samples appearing slightly more frequently in the leading positions.

Französisch Buchholz leads the graph, followed by Kladow with the strongest underprediction. Very few districts seem to be associated with underpredicting bias, such as Prenzlauer Berg, Wannsee and Friedrichshagen while there are many districts associated with considerable overpredicting bias, with the strongest bias being Hermsdorf, Marienfelde, Marzahn and Altglienicke.

When it comes to the districts with the widest residual distribution, most of them seem to correspond to the districts with the highest overall errors and big error outliers, like Prenzlauer Berg, Friedrichshagen, Charlottenburg, Mitte, Wilmersdorf and Weißensee.

property_condition	n	MAE	RMSE	R2	bias
Old Building	15	148.6	274.5	1.0	-20.6
Renovated	271	141.8	231.1	0.9	-14.9
Unknown	1837	136.2	212.3	0.9	12.3
Partially Renovated	5	122.9	176.5	0.5	-48.4
Like New	322	119.7	216.1	0.9	-19.9
New / First Occupancy	370	95.8	166.9	0.9	-2.7
Well Maintained	288	79.3	150.0	0.9	25.0
Other	5	77.4	144.7	1.0	-56.5
Needs Renovation	5	45.0	61.6	1.0	45.0

Table 2: Metrics for each value of 'property_condition' feature

By breaking down the testing sets into mini sets based on values for certain features, we can evaluate if certain values are noisier or complicated than the others.

Table 2 breaks down the performance for the condition of the apartment. Most of the values seem to have a very good R^2 except 'Partially Renovated', that means that our model is not good at explaining the variance of this category. One possible reason is that partially renovated properties can differ substantially in their actual condition, making the category less homogeneous. In terms of RMSE, most categories show errors of roughly €200, while Other and Needs Renovation have lower errors. However, the lower RMSE in these categories should be interpreted carefully, as it may partly result from their smaller number of observations or lower variability in rent. 'New / First Occupancy' has an almost negligible bias of -€2.7, indicating that the model is, on average, neither substantially overestimating nor underestimating rents for this category.

Subgroup metrics for categories with very small sample sizes should be interpreted cautiously because estimates such as R^2 and RMSE can be highly unstable with few observations.

heating_type	n	MAE	RMSE	R2	bias
Unknown	1322	146.6	214.0	0.9	5.8
Central Heating	900	123.7	216.6	0.9	10.7
Unit Heating	157	96.0	164.4	0.9	25.6
Underfloor Heating	674	94.1	181.8	0.9	-3.1
Individual	65	82.8	138.5	1.0	-23.4

Table 3: Metrics for each value of 'heating_type' feature

Table 3 refers to heating type. The performance of the model is relatively consistent across heating types, with similar R^2 and RMSE values. This suggests that heating type itself does not strongly determine how accurately the model predicts rent. Bias is making the whole difference, with categories like 'Unit Heating' and 'Individual' having much higher bias than the rest, indicating that the model tends to systematically overestimate or underestimate rents for these properties. This could suggest that the effect of these heating systems is not fully captured by the model, potentially because these categories interact with other characteristics such as building age, property condition, or energy efficiency.

rooms	n	MAE	RMSE	R2	bias
6.0	8	419.0	673.2	0.7	-189.7
4.5	12	222.0	289.8	0.8	-113.8
5.0	76	211.2	334.6	0.9	-24.2
4.0	314	145.5	244.4	0.9	-12.7
3.0	791	145.1	233.4	0.9	-5.7
3.5	28	117.7	205.7	0.7	-14.0
2.5	103	117.1	191.4	0.9	5.8
2.0	1197	113.8	173.7	0.9	10.8
1.0	536	91.5	144.6	0.9	36.3
1.5	49	79.4	101.9	0.9	10.0

Table 4: Metrics for each value of 'rooms' feature

For number of rooms there is a clear relationship that can be observed between the number of rooms and prediction error. Apartments with a higher number of rooms, like 4 to 6 rooms, tend to have a much larger RMSE and bias. This could be since most of the outliers were apartments that were with very large areas and a number of rooms.

8 Model Explainability

Understanding model behavior is very important because we need to understand on what the model bases its decision. It is necessary to know that the model has learned the right patterns that align with economic interpretation of the rental market. We should expect the model to favor features such as area, number of rooms, location and valuable amenities for its evaluation.

8.1 Feature Importance

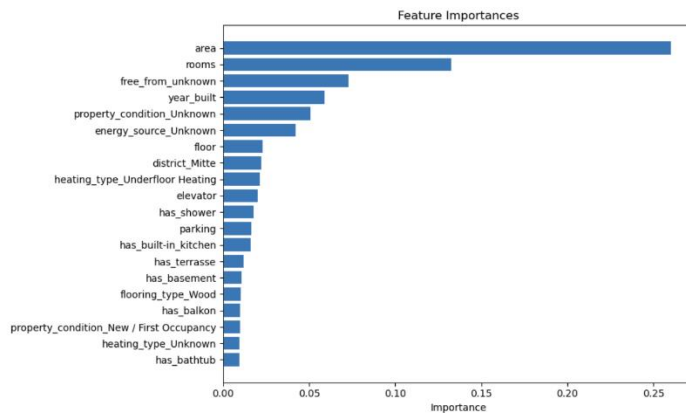


Figure 38: Feature importance ranking

Feature importance shows which feature impacts the final prediction the most. Area has the strongest impact with 26.6%, followed by number of rooms, availability (free_from_unknown), year built, floor, district (Mitte). Many Unknown categories make it in top 15.

8.2 SHAP Values

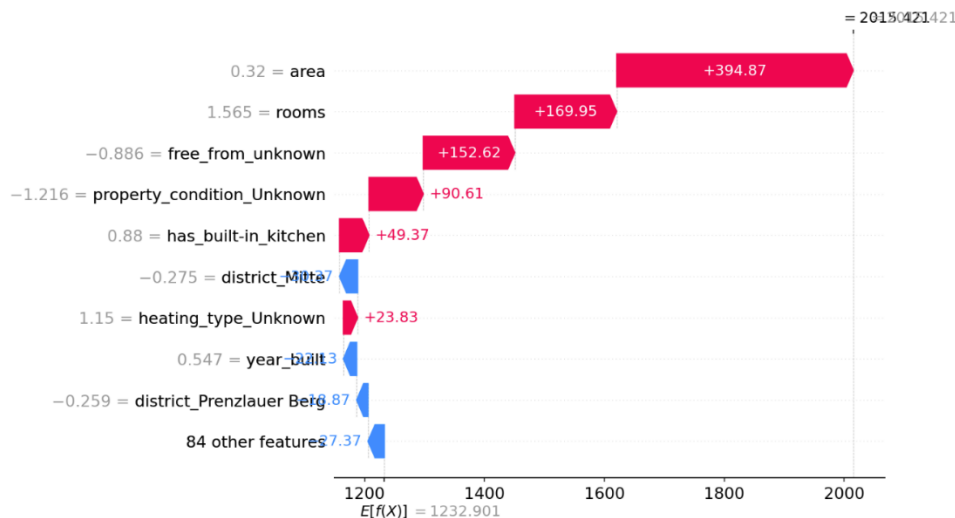


Figure 39: SHAP values for overall prediction

The strongest positive contributor is area. Number of rooms has also a strong positive effect.

Availability feature contributes twice positively. Unknown values, especially energy source and property condition have a negative effect. When it comes to districts, 'Mitte' has the strongest effect and its absence is associated with lower rent prices. Underfloor heating and built-in kitchen have a positive association with the prices.

8.3 Partial Dependence

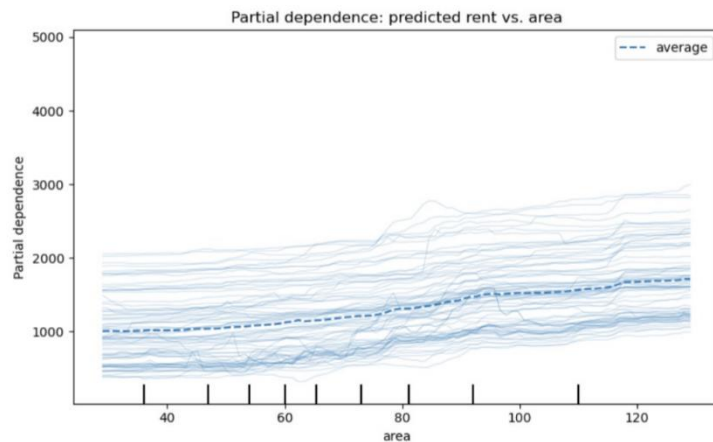


Figure 40: Partial dependence of predicted rent and area

Partial dependence shows the average predicted rent as the value of a feature is varied while averaging over the observed values of the other features. In this case, the plot shows a gradual increase in predicted rent as apartment area increases. A gradual increase in predicted rent is clearly visible and there is more noise for 45–70 m² and 70–100 m² due to diverse property mix.

8.4 Individual Prediction Breakdown

- Example: Property #27, actual 1920 €, predicted 1909 €.
- Tree predictions ranged from 0 to 3200 €; 113 trees in correct range.
- SHAP waterfall: area +436 €, rooms +227 €, immediate move-in positive effect, unknown condition negative effect.

9 Ethical Considerations

While collecting and processing data it is very important to consider ethical and legal considerations to have a complete and proper data science project.

9.1 GDPR & Privacy

- No PII collected.
- Full location generalized via k-anonymity before publication.

9.2 K-Anonymity Check

- District alone produced a minimum equivalence-class size of 24.
- District + rooms: 40 unique rows (0.26%).
- District + rooms + floor: 912 unique (5.84%).

- District + rooms + floor + year_built: 26.14% unique.
- Worst case (district + rooms + floor + area_bin + year_built): 42.38% unique.
- Recommendation: bucket year_built into decades and group floor into ranges before publication.

9.3 Bias Discussion

- **Historical Bias:** Data reflects historical rent patterns.
- **Population Bias:** Listings may not represent all apartments.
- **Self-Selection Bias:** Landlords who list online may differ.
- **Temporal Bias:** Single scraping window.
- **Measurement Bias:** Scraper failures.
- **Omitted Variable Bias:** Missing features like exact condition, interior quality.
- **Representation Bias:** Some districts over/underrepresented.
- **Aggregation Bias:** Simpson's Paradox possible.
- **Evaluation Bias:** Metrics may not reflect all subgroups equally.

9.4 Fairness

The dataset does not contain explicit protected attributes such as race or gender. Therefore, direct subgroup fairness analysis based on these attributes could not be performed. However, the absence of protected attributes does not eliminate the possibility of unequal model performance across groups or indirect effects through proxy variables. In this project, subgroup performance was therefore examined using available characteristics such as district, property condition, heating type and number of rooms.

10 Limitations & Future Work

Like every project, there are always limitations and challenges that can be encountered. Some of them include the fact that the data was scraped into a single period, which doesn't permit for time-series analysis. There were many missing values for some features. The final model, although achieving a very good performance, still under- or over-predicts extremes.

Future work could include more collection of data longitudinal data, derivation of more information out of current features, for example out of location features a distance to center or neighborhood quality could be derived. Deployment as API for real-time predictions could also be beneficial to users.

11 Conclusion

This project developed and evaluated a machine learning model for predicting cold rent for apartments in Berlin. Multiple regression approaches were compared, with XGBoost and Random Forest achieving the strongest predictive performance. Random Forest was selected as the final model based not only on predictive performance but also on its suitability for the project's uncertainty-analysis objective.

The model demonstrated strong predictive performance with an R^2 of 0.9198 on the held-out test set, while the analysis of prediction errors showed that performance decreases for particularly expensive or otherwise unusual properties. Tree disagreement was found to be associated with absolute prediction error and was therefore used as an empirical uncertainty indicator. However, this measure should not be interpreted as a calibrated confidence interval.

The project also identified important limitations related to the single data-collection period, missing property information, potential representation bias, and privacy risks associated with combinations of quasi-identifiers. Overall, the project demonstrates an end-to-end machine learning workflow from web-based data collection to model deployment and explainability. Future work should focus on longitudinal data collection, richer geographical features and more rigorous uncertainty calibration.

12 Appendices

- **Notebooks:** rent_prices_cleaning.ipynb, rent_prices_consumption.ipynb, rent_prices_modeling.ipynb, rent_prices_explainability.ipynb
- **Dataset:** Not published yet.
- **Model Card:** model_card.pdf
- **Visualizations:** See graphs/ folder for all figures.
- **Code:** Available on GitHub.